

Verifiable RL Has an Information Budget

AlphaZero renews its 50/50 frontier. Fixed LLM math datasets run out of useful middle-band signal. RL then sharpens a broad rollout distribution into problem seeds.

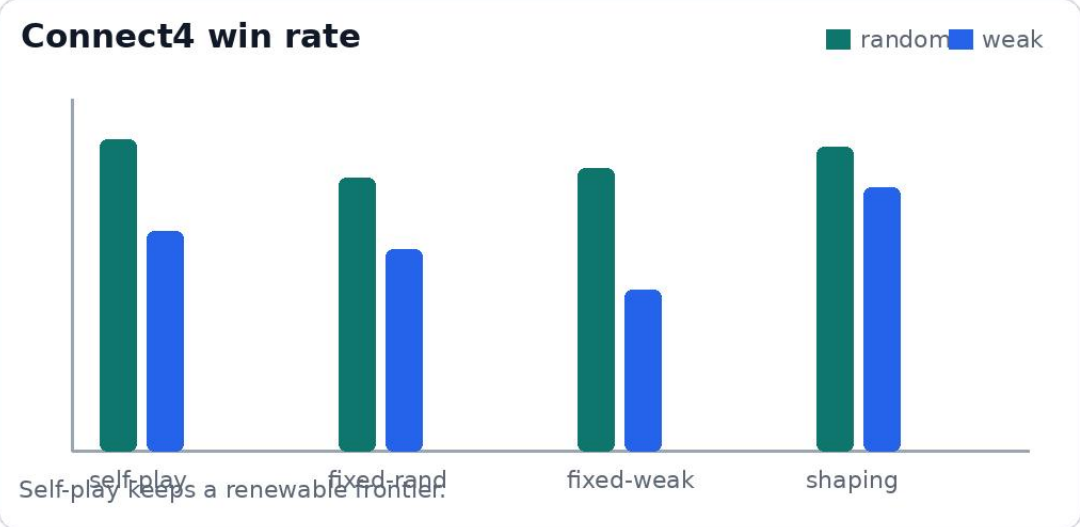
One-line thesis

The unit is a rollout/path information integral. AlphaZero can keep adding information through self-play; fixed-prompt LLM RLVR mostly spends a finite verifier source; transfer geometry tells us which prompt seeds interact.

1. Traditional RL: renewable information

Self-play keeps the policy near opponents and positions at its current level. Outcomes stay near half win / half loss, so the cumulative information integral can grow with training time.

SELF-PLAY VS RANDOM SELF-PLAY VS WEAK SHAPED VS WEAK
0.886 **0.625** **0.750**



2. LLM math: finite middle band

Math prompts often become all-right or all-wrong. For each prompt, useful Bernoulli-verifier information is finite; over the dataset, interaction directions make the joint useful cap smaller than a naive sum.

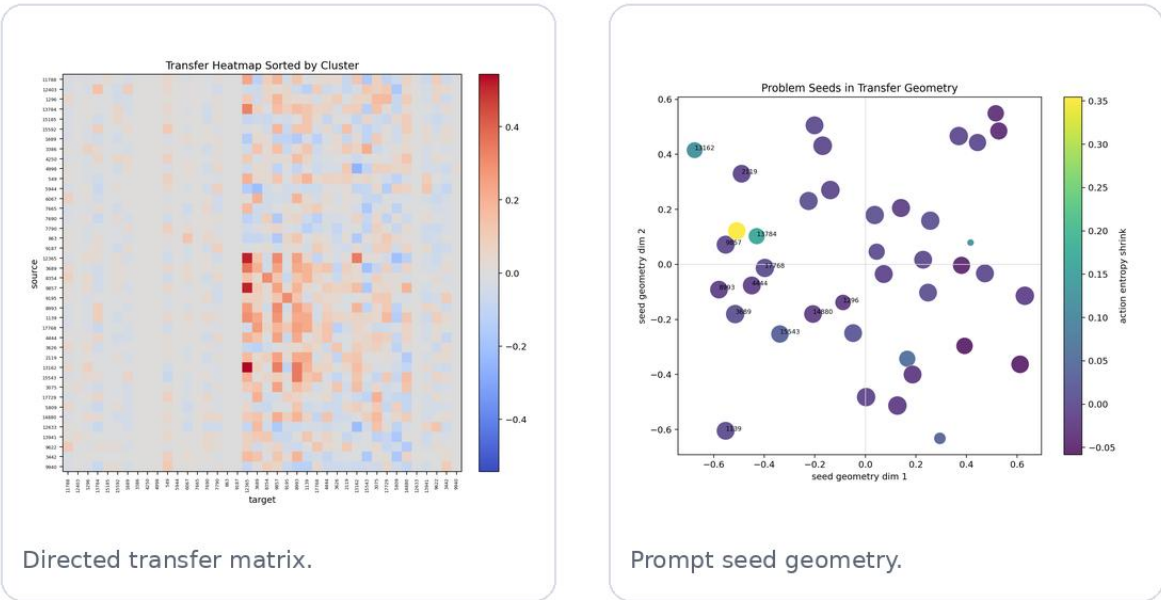
USED PROJECTION DIAG DELTAP OFFDIAG / DIAG
41.5% **0.0706** **0.129**



3. RL sharpens: watermelon -> seeds -> geometry

After RL, a broad rollout distribution becomes lower-entropy problem-conditioned seeds. Train on source i , evaluate target j , and the transfer matrix becomes a map of dataset distance.

```
P_i = sum_t H_b(p_i(t))
G_i = sum_t H_b(p_i(t)) * Delta p_i^+(t)
T_i = [DeltaP_i1, ..., DeltaP_iN]
d(i,j) = 1 - cos(T_i, T_j)
```



Supporting contrast: diffusion reward channels

Diffusion RL is not binary math verification, but it shows the same principle: more reward signal only helps when it distinguishes rollouts and aligns with the target evaluator. In the 10-prompt SDXL DDPO run, IR+PS has the richest active feedback diagnostic, while held-out HPSv2 is controlled by alignment.

IR+PS I_ACC PS HPSV2 Z IR+PS HPSV2 Z
0.9335 **-0.0947** **-0.0958**



Poster takeaway

To make LLM math RL more AlphaZero-like, do not only optimize harder. Renew the frontier, allocate rollouts to high-information prompts, enrich feedback, and use transfer geometry to avoid redundant data.